

Lab Session 7 (Open-Ended Lab assignment)	
Open Ended Lab / Project	
Blooms Taxonomy	GAs
P2 C3 A3	GA- 2 GA- 5 GA- 6
Syed Mesum Ali Shah	76162

Title:

Mini Company Management System (C Language - Console Based)

Motivation:

The main idea of this project is to create a simple system that can manage basic work of a small company. In many small setups, tasks, employee records, and money details are handled manually, which can cause mistakes and confusion. Implementation of additional modules like Attendance Management, Task Status Tracking, and Salary Calculation based on attendance

Problem Statement:

In many small companies, daily work like managing employees, assigning tasks, and handling financial records is done manually. This can lead to mistakes, missing data, and lack of proper organization.

There is no simple system to manage users based on roles like Admin, Accountant, and Employee in an organized way. Because of this, it becomes difficult to track tasks, manage salaries, and keep records safely.

Methodology:

- Proposal Submission and Planning
- Understanding system requirements (Admin, Accountant, Employee) Design of Data Structures (1D Array, 2D Array, Pointers)
- Designing simple data structures using arrays, structs, and file handling
- Implementation of this project in C-Language
- Testing each module (login, task assign, transactions)

- Final integration of all modules into one system

Design / Ways & Means:

- System will be console-based application in C language
- Role-based design (Admin, Accountant, Employee)
- File handling will be used to store and retrieve data
- Simple menus will be used for user interaction
- Login system will control access based on username and password
- Structs and arrays will be used to organize data in program

System Enhancements:

- Attendance management system is added where employees mark daily attendance
- Task management is enhanced with status (Pending, In Progress, Completed) and deadline
- Salary calculation is based on attendance records instead of fixed amount
- Search functionality is added for users and transactions
- Modular programming approach is used for better structure and readability

Data Handling:

- The system uses multiple text files for data storage:
- users.csv or txt (stores login information and roles)
- tasks.csv or txt (stores task details with status and deadlines)
- attendance.csv or txt (stores daily attendance records)
- transactions.csv or txt (stores financial data)

This ensures data persistence and improves system reliability.

Conclusion:

This project helps in understanding how a simple system can manage basic operations of a small company in an organized way. It reduces manual work by handling users, tasks, attendance, and financial records through a computer-based system.

System Flow:

